Why New Customers Stopped Coming Back

Diagnosing a hidden decline in repeat purchase — and what to do about it

E-commerce order log, Jan 2024 – Jun 2025 (simulated data) · github.com/Newnew-niu/ecommerce-repeat-purchase-analysis

The reported KPI looked stable while new-customer repeat fell by 16%

−2.0pp

Headline 30-day repeat

58.2% → 56.3%. What the business reported

−7.5pp

New-customer 30-day repeat

46.2% → 38.7% ($p < 0.001$)

−7.6pp

Cohort M+1 retention

34.7% → 27.2%, cohorts since Nov 2024

1. **Support tickets:** Repeat collapses at the 3rd ticket (odds ratio 0.22); orders with a ticket rose from 14% to 27%

2. **Electronics & Home:** Highest order values, lowest repeat; tickets per order up 2.9x and 1.7x

3. **Coupon dependency:** Customers who habitually buy on coupon repeat ~15pp less; a single coupon has no measurable effect

Baseline = 2024Q2–Q3, recent = 2025Q1–Q2. Sources: outputs/tables/05_kpi_summary.csv, 02_cohort_retention.csv

Repeat customers drive most revenue, so a retention leak is expensive

80%

of GMV comes from repeat customers, who are 54% of customers

They spend less per order ($87 vs $111) but order 4.7 times on average. Their value comes from coming back, which makes the first-to-second order step the key lever.

Questions this analysis answers

1. Is repeat purchase really declining, or is it noise?
2. Which customers are affected: new or established?
3. What is associated with the decline: service, category, coupons or channel?
4. Which actions are worth funding, and how do we prove they work?

Repeat customer = 2+ orders in the observation window. Source: outputs/tables/05_customer_type_comparison.csv

Approach: clean, measure without bias, explain, then size and test

1

Data

42,776 raw order rows, ~14,000 customers, 9 fields; 10 realistic data-quality issues

2

Clean

97.7% of rows kept; every removal counted; last 30 days censored

3

Measure

Order-, customer- and cohort-level KPIs; segment profiles

4

Explain

Logistic regression (user-clustered SEs) + gradient boosting, forward-in-time holdout AUC 0.72

5

Size & check

Break-even business cases, sensitivity grid, SQL cross-check (9/9 pass)

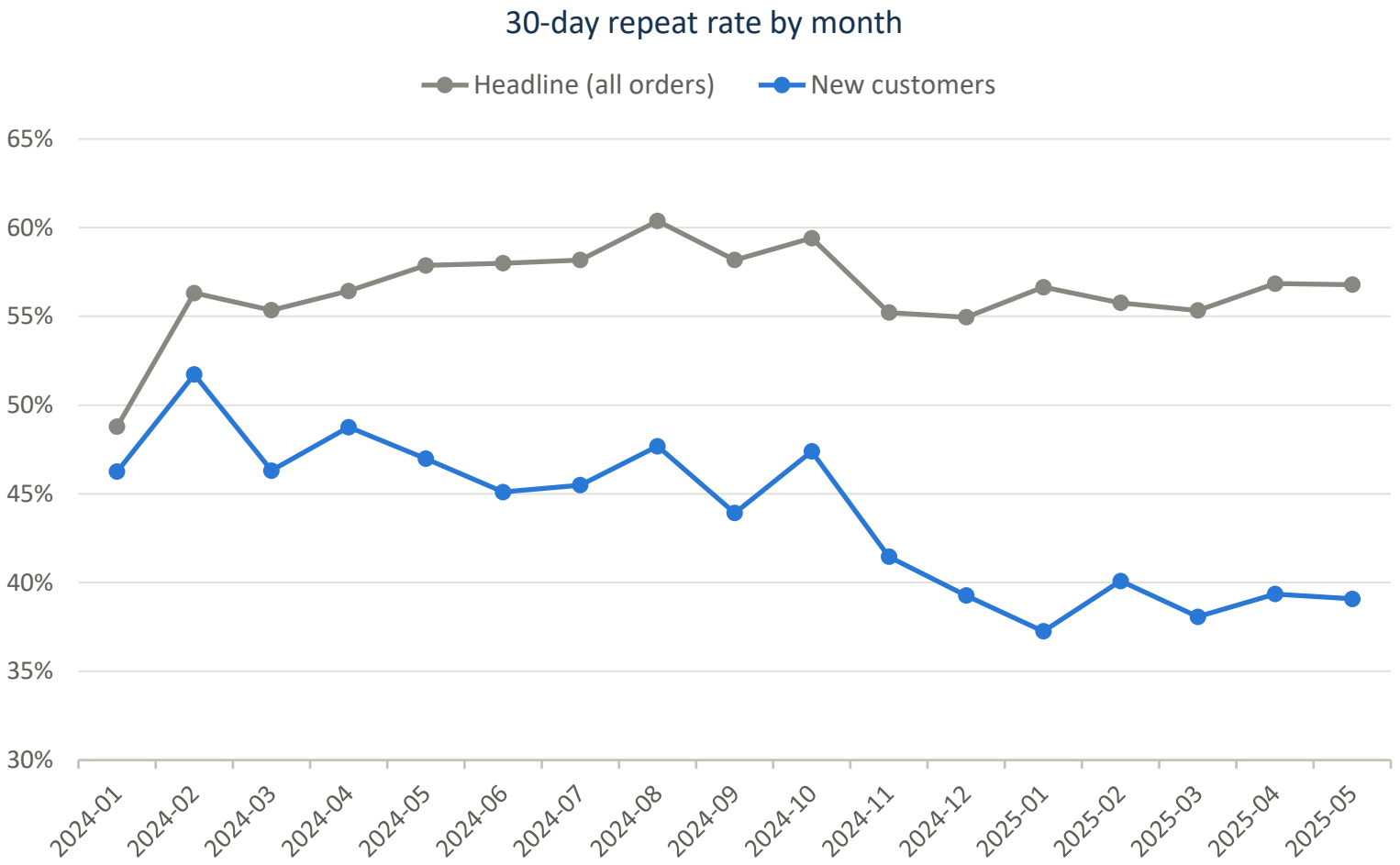
Data provenance. No production data was available, so the order log is simulated with known patterns. The analysis never reads the simulator's settings; it rediscovers them from the data. This deck therefore demonstrates analytical method. The numbers are not real-world facts.

Three repeat metrics, three different questions

KPI	Unit	Numerator ÷ denominator	Use it for
Order-level 30-day repeat	Order	Orders followed by another order ≤30 days ÷ orders ≥30 days before snapshot	What was reported; distorted by customer mix
New-customer 30-day repeat	Customer	First-time customers who reorder ≤30 days ÷ first-time customers ≥30 days before snapshot	Primary KPI for acquisition quality
Cohort M+1 retention	Customer × month	Cohort customers ordering in the next calendar month ÷ cohort size	Retention trend by acquisition month
Average order value	Order	GMV ÷ orders	Basket value; compare segments

Why it matters: repeat rate, retention rate and cohort retention use different units and denominators. Mixing them up is how this decline went unnoticed.

The headline rate hid the decline as loyal customers grew as a share of orders



42%

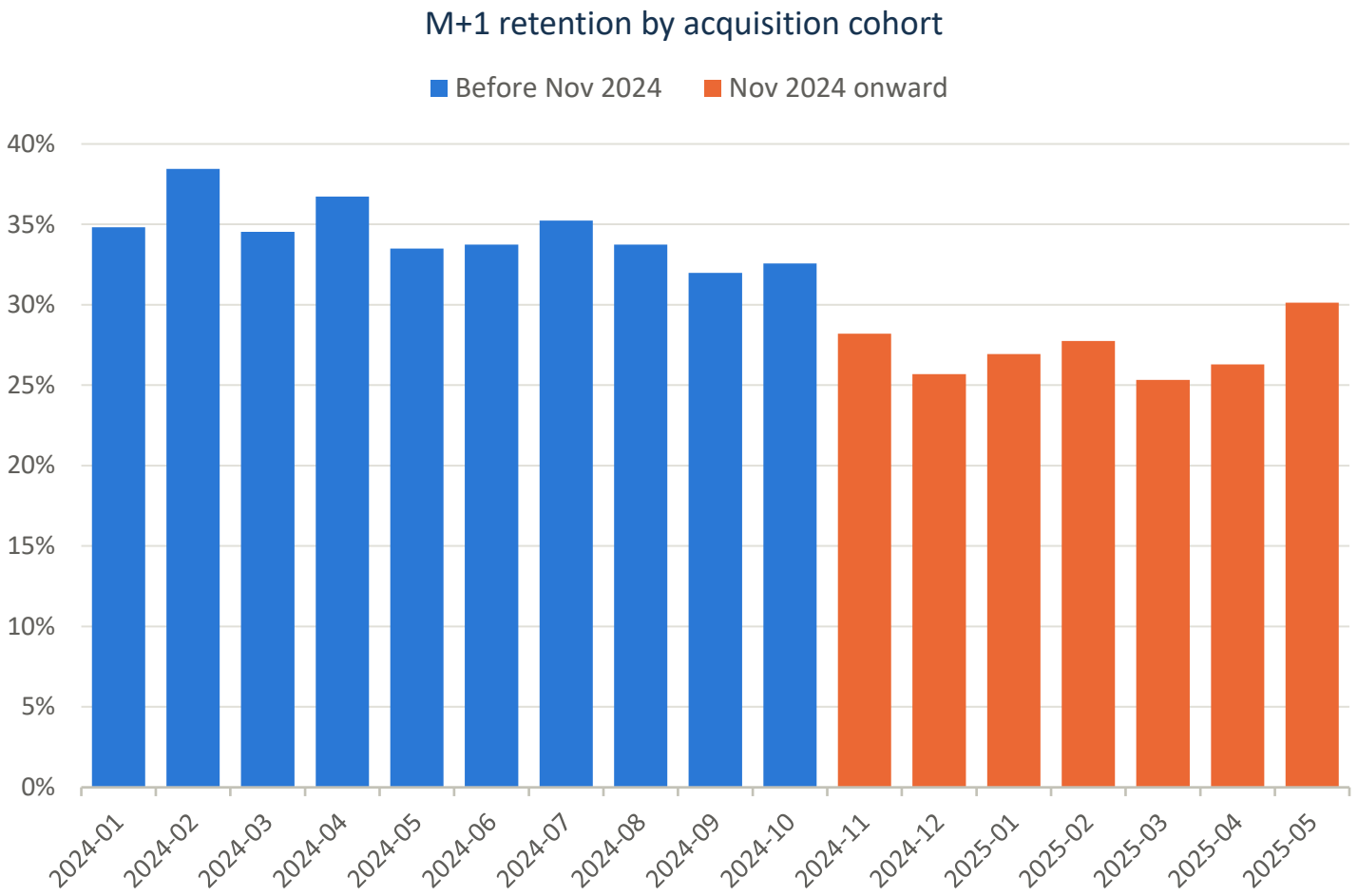
of orders from loyal customers

Customers with 4+ prior orders, May 2025 (0% in Jan 2024). They repeat at ~70%.

So the average barely moved, while each new cohort came back less often. Track new-customer repeat and cohort retention, not the blended headline.

Source: outputs/tables/02_monthly_trend.csv, 02_order_mix_drift.csv · Orders within 30 days of the snapshot excluded

Every cohort acquired since November 2024 retains worse, at any time window



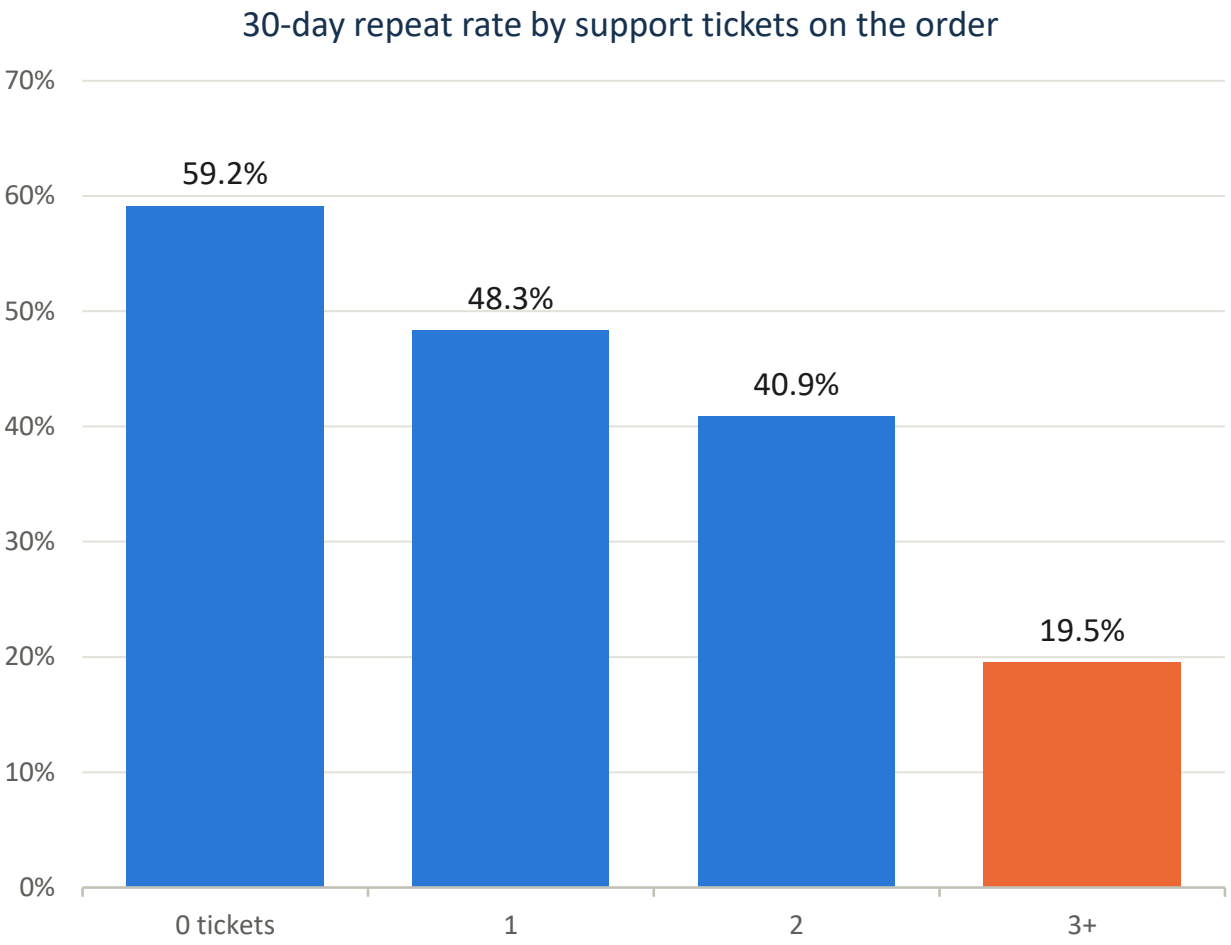
Is it just the 30-day window?

Window	Baseline	Recent	Change
30 days	46.2%	39.5%	**–6.8pp**
60 days	49.9%	42.9%	**–7.0pp**
90 days	54.2%	46.4%	**–7.8pp**

No. The gap holds at 30, 60 and 90 days (all p < 1e-10). New-customer repeat, share of customers who order again. Baseline cohorts Apr–Sep 2024 (n=4,932); recent Nov 2024–Mar 2025 (n=4,195).

Source: outputs/tables/02_cohort_retention.csv, 05_window_sensitivity.csv · Robustness uses the same customers under each window

Repeat purchase collapses at the third support ticket, and ticket load has doubled



0.22

Odds ratio, 3+ tickets

95% CI 0.16–0.30, after controls

14%→27%

Orders with a ticket

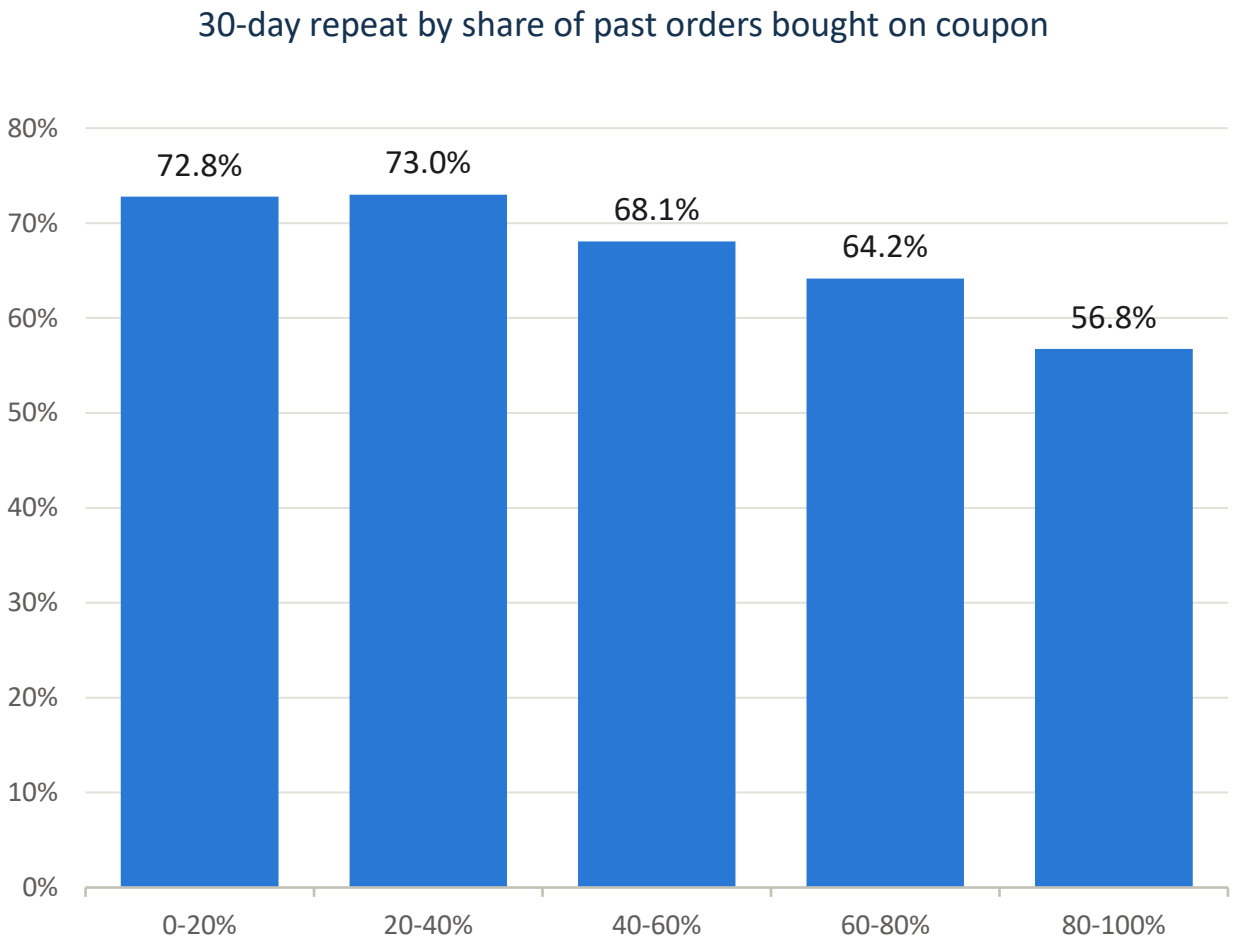
Jan 2024 → Jun 2025

Where the strain landed: Electronics tickets per order rose from 0.24 to 0.69, Home from 0.25 to 0.42. These are the highest-value categories. The ticket gradient holds within every category.

Source: outputs/tables/05_ticket_robustness.csv, 03_odds_ratios.csv, 02_ticket_trend.csv, 02_ticket_by_category_month.csv

Simulated data · portfolio project

Coupons are not the problem; coupon dependency is



0.96

Coupon on this order

Not significant (p = 0.28)

0.50

Habitual coupon use

Always vs never; p < 0.001

Implication: do not stop couponing. Break the habit for customers who only buy on discount. This is also the weakest causal link (low-intent customers may seek coupons), so test it before rollout.

Customers with 3+ prior orders only, to avoid one-order customers having a 100% coupon rate by construction. Source: 02_tipping_discount.csv, 03_odds_ratios.csv

Simulated data · portfolio project

Service and coupon dependency explain about half of the decline; the rest needs new data

Share of the controlled decline in 30-day repeat (log-odds scale)



Hidden by mix

Measured naively the period effect is not significant (OR 0.96, p = 0.10). With controls it is clear (OR 0.77).

Partly explained

Tickets, returns and coupon dependency absorb 49% of it. These are the levers in the recommendations.

Next data to collect

Delivery times, stockouts and assortment changes. The order log cannot see them, and they are the best candidates for the rest.

Logit ladder on the post-Oct-2024 period effect, with lifecycle, category, channel and order-value controls. Source: 03_mediation_ladder.csv

Four actions, ordered by strength of evidence and value

1 Switch the KPI

This week · no cost

Report new-customer repeat and cohort M+1 weekly; keep the headline as a secondary metric.

Measure: adopted in the weekly business review

2 Fix Electronics & Home after-sales

Base case $172k net / yr

Product information, proactive delivery updates and inbound quality checks to cut tickets at the source.

Measure: tickets per order ≤ 0.35; staged rollout by category

3 Recover at the 2nd ticket

Base case $50k net / yr

Automated recovery at 2 tickets; senior-agent callback at 3+. Tiering follows from break-even costs.

Measure: 2 tickets: 10% holdout, fewer escalations; 3+: 50/50 test, repeat

4 Test coupon-habit reduction

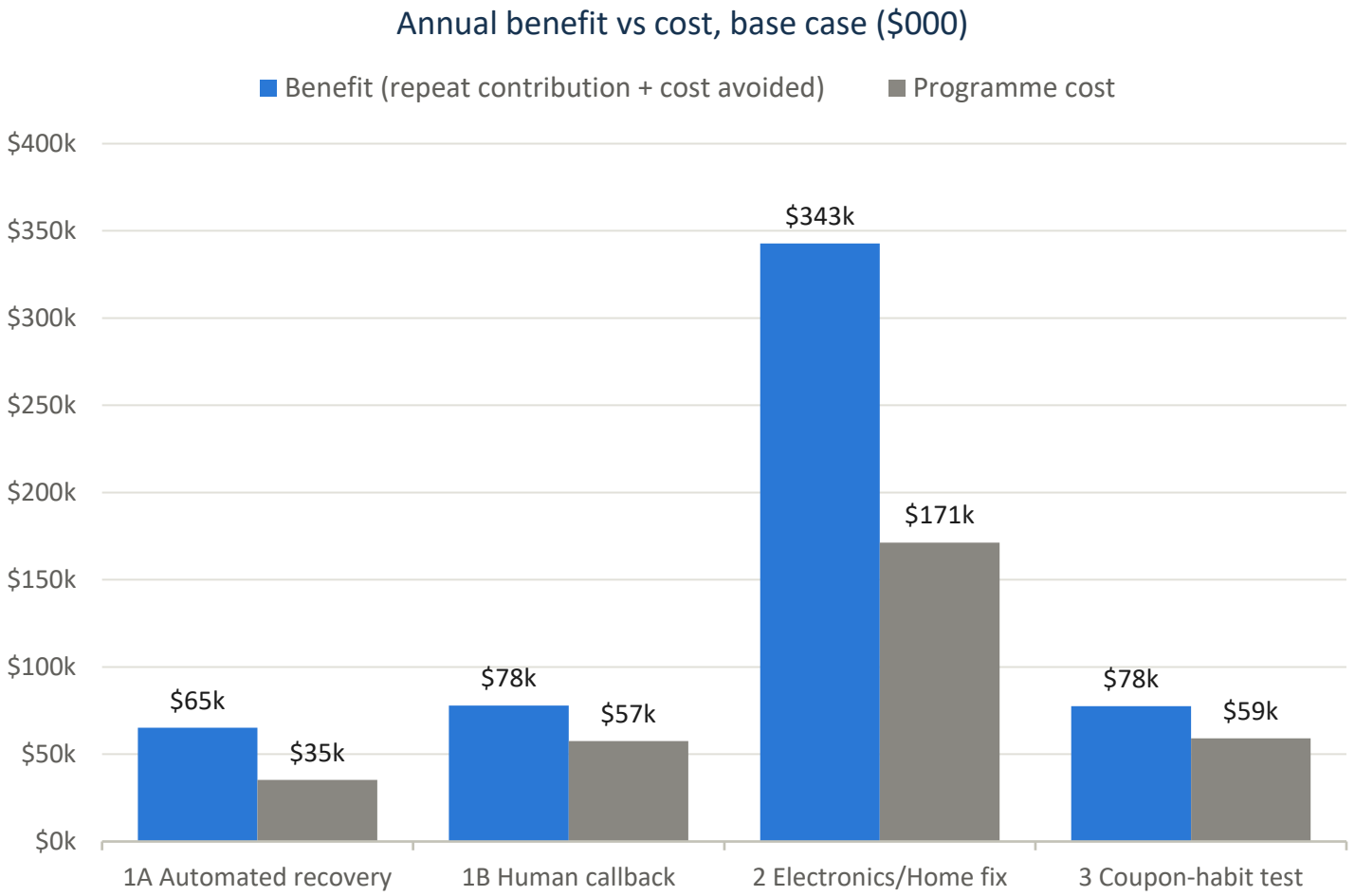
Base case $18k net / yr if proven

Targeted CRM for customers who buy ≥60% on coupon. Weakest causal evidence.

Measure: A/B test before any rollout; repeat and margin

Net contribution = base-case annual planning estimate (10% sample assumption). Source: outputs/tables/04_business_cases.csv

Base case: ~$240k net contribution a year, but only if the uplift is proven



$240,321

Net contribution / yr

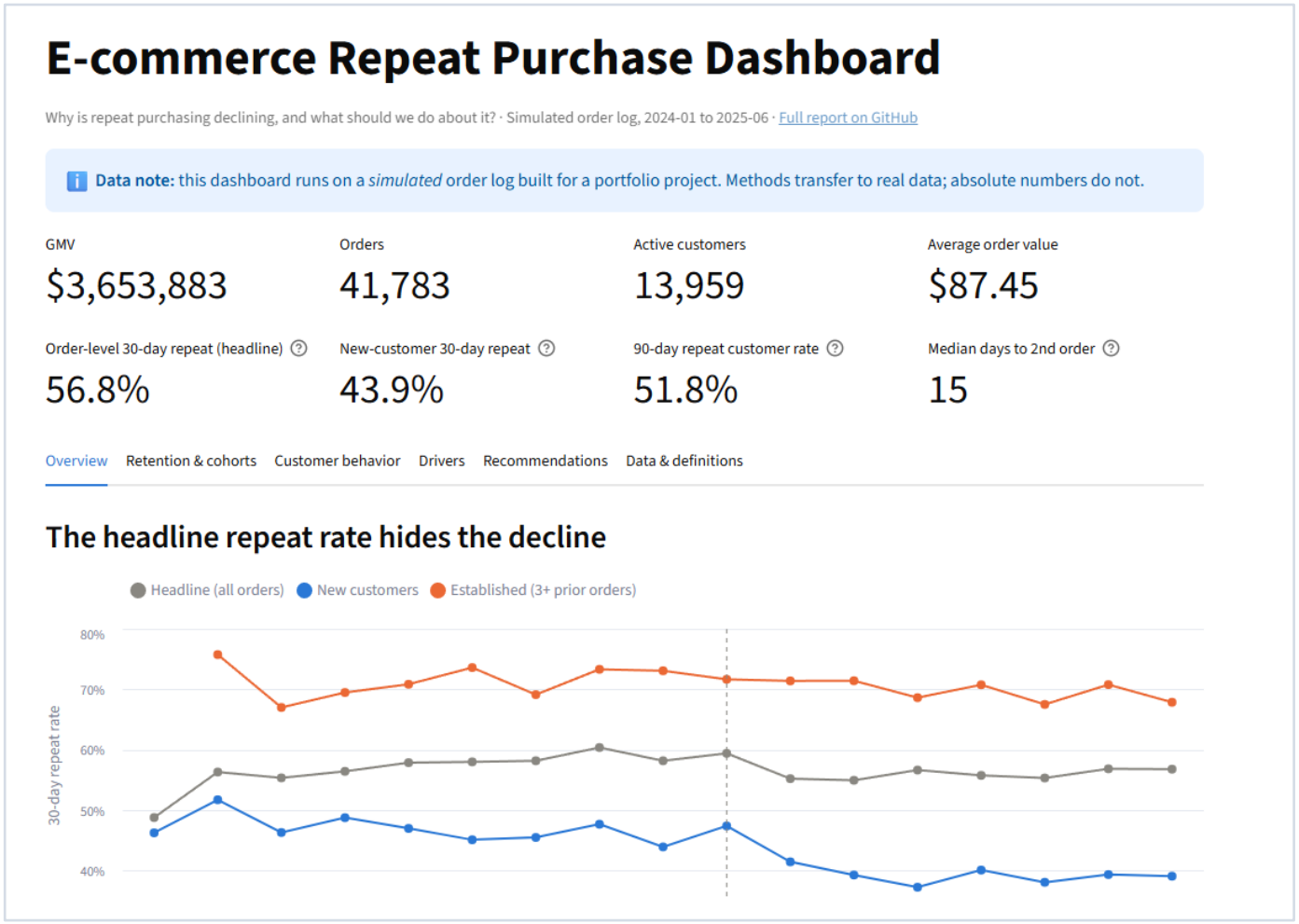
on $323,170 cost · net ROI 0.74x · $852,230 incremental GMV

Risks, stated up front

- 13 of 18 sensitivity scenarios are value-positive.
- With zero repeat uplift the portfolio loses $117,615; savings alone do not pay for it.
- Effects come from observational models, so run the controlled tests first.

Annualised, contribution margin 42%, assumes the data is a 10% customer sample. Source: 04_business_cases.csv, 04_sensitivity.csv

An interactive dashboard keeps the right KPIs in front of the team



- KPI tiles use the same code as the analysis (src/kpis.py); automated tests check they match
- Filters: month range, acquisition channel, category
- Tabs: overview, cohorts, customer behaviour, drivers, recommendations, definitions
- Every chart has hover tooltips and a plain-English caption

Screenshot of dashboard/app.py running locally (Streamlit). Run: streamlit run dashboard/app.py

Roadmap: change the metric this week, prove the levers within six months

1

Week 1–2

Switch reporting to new-customer repeat & cohort M+1; real-time alert at 2 tickets

Analytics, Data Eng

2

Week 3–8

Launch automated recovery (10% holdout) and callbacks (50/50 test, ~6 months)

Customer Service Ops

3

Month 2–5

Electronics/Home: product content, delivery updates, inbound QC; staged rollout

Merchandising, Fulfilment

4

Month 3–6

Randomised test of coupon-habit reduction; scale only on a significant result

CRM, Analytics

5

Month 4+

Instrument delivery time, stockouts and assortment to explain the remaining ~50%

Data Eng

Owners are suggested functions. Dates are relative to approval.

What this shows, and what it does not

Conclusion

- The repeat decline is real, concentrated in new customers, and was hidden by the KPI definition.
- Support tickets, Electronics/Home after-sales and coupon habits are the measurable levers.
- Change the metric now; fund the levers through controlled tests.

Limitations

- Simulated data: methods transfer, numbers do not
- Observational models; no experiment has been run
- About half of the decline is unexplained by the available fields
- Dollar figures rely on a 10% sample assumption and planning costs
- Ticket and return timing may overlap the 30-day outcome window

Full report, code and dashboard: github.com/Newnew-niu/ecommerce-repeat-purchase-analysis